

Lab 3

Constructors

Constructor:

A constructor is a special type of member function that is called automatically when an object is created. It has the following properties:

- has the same name as the class.
- does not have a return type, and
- is public.

For Example:

```
class Wall {  
    public:  
        // create a constructor  
        Wall() {  
            // code  
        }  
};
```

C++ Default Constructor

A constructor with no parameters is known as a **default constructor**.

Example 1:

```
#include<iostream>  
using namespace std;  
  
// declare a class  
class Wall {  
    private:  
        double length;  
  
    public:  
        // default constructor to initialize variable  
        Wall()  
        {  
            length=5.5;  
            cout << "Creating a wall." << endl;  
            cout << "Length = " << length << endl;  
        }  
};
```

```
int main() {  
    Wall wall1;  
    return 0;  
}
```

C++ Parameterized Constructor

In C++, a constructor with parameters is known as a parameterized constructor. This is the preferred method to initialize member data.

Example 2:

```
#include <iostream>  
using namespace std;  
  
// declare a class  
class Wall {  
private:  
    double length;  
    double height;  
  
public:  
    // parameterized constructor to initialize variables  
    Wall(double len, double hgt)  
    {  
        length=len;  
        height=hgt;  
    }  
  
    double calculateArea() {  
        return length * height;  
    }  
};  
  
int main() {  
    // create object and initialize data members  
    Wall wall1(10.5, 8.6);  
    Wall wall2(8.5, 6.3);  
  
    cout << "Area of Wall 1: " << wall1.calculateArea() << endl;  
    cout << "Area of Wall 2: " << wall2.calculateArea();  
  
    return 0;  
}
```